higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS
(Second Paper)
NQF LEVEL 3

(10501053)

22 February 2019 (X-Paper)
09:00 –12:00

A nonprogrammable scientific calculator may be used.

This question paper consists of 9 pages, 2 formula sheets and 1 graph paper.

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TIME: 3 HOURS
MARKS: 100

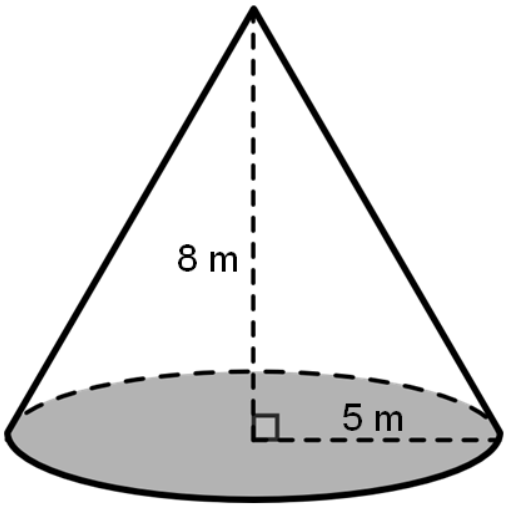
INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Clearly show ALL calculations, diagrams, graphs, et cetera which you have used in determining the answers.
5. If necessary, answers should be rounded off to THREE decimal places, unless stated otherwise.
6. Diagrams are NOT drawn to scale.
7. Write neatly and legibly.

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QUESTION 1

1.1 The following figure represents a cone. The radius of the base is 5 m and the perpendicular height is 8 m.



Calculate:

1.1.1 The volume of the cone (2)

1.1.2 The area of the curved surface of the cone

HINT: Calculate the slant height first (3)

1.2 A rectangular-shaped glass container is filled to the brim with water. The container has a base measuring 20 cm by 10 cm and a height of 15 cm.

Spherical-shaped glass marbles, each with a diameter of 4 cm are then placed in the container.





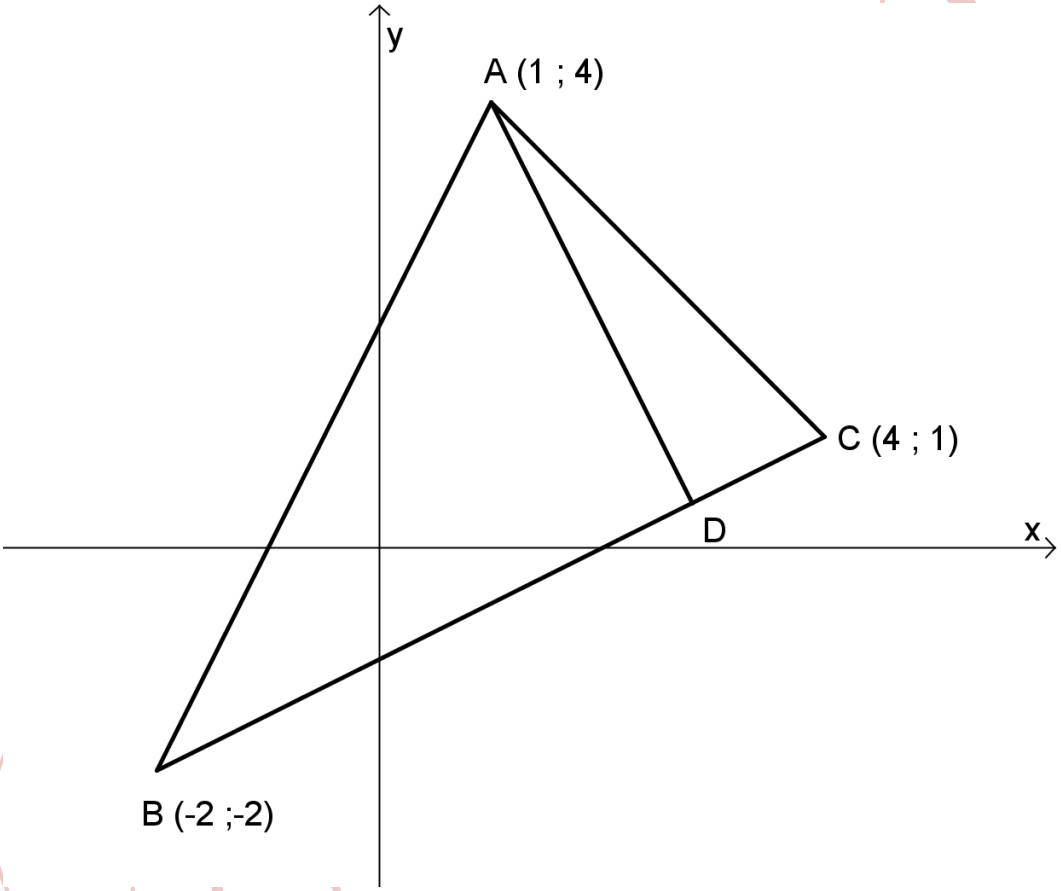
Calculate:

1.2.1 The volume of a marble (2)

1.2.2 The volume of the water in the container before the marbles were placed in it (2)

1.2.3 The number of marbles (to the nearest whole) that were placed in the container if the amount of water that remained in the container after the marbles were removed, was 1 000 cm^3 (3)

1.3 In the diagram, ΔABC has vertices A(1; 4); B(-2;-2) and C(4; 1).



Determine:

1.3.1 The length of AB (3)

1.3.2 The coordinates of the M; the midpoint of AB (2)

1.3.3 The equation of AD if $AD \perp BC$ (5)

1.3.4 The inclination of AD (3)

[25]

QUESTION 2

2.1 Calculate, without the use of a calculator, the value of:

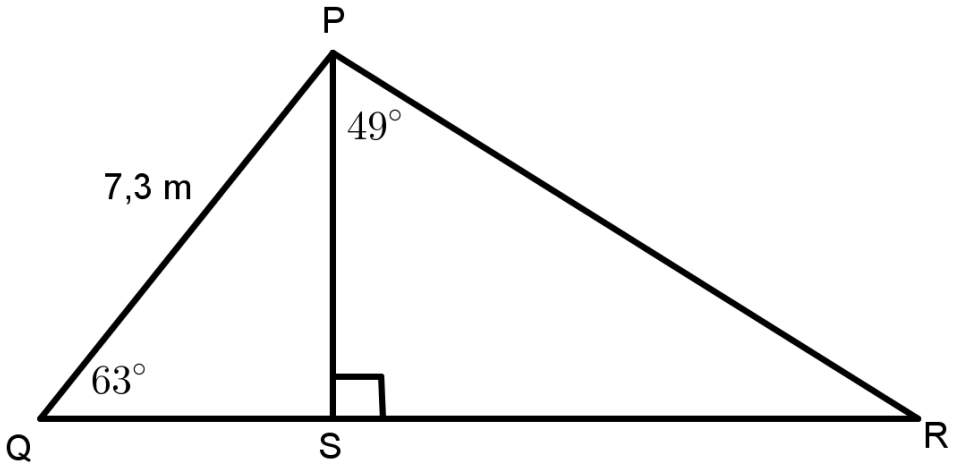
$$\frac{\sin^2 240°}{\cos^2 120°} + \sin 330°.\tan 225°$$ (4)

2.2 Make use of trigonometric identities to prove the following:

$$\frac{8}{\sin^2 A} - \frac{4}{1+\cos A} = \frac{4}{1-\cos A}$$ (5)

2.3 Calculate the value/s of θ if $2\tan(\theta + 10°) = 4{,}162$ where $\theta \in [0°; 360°]$. (5)

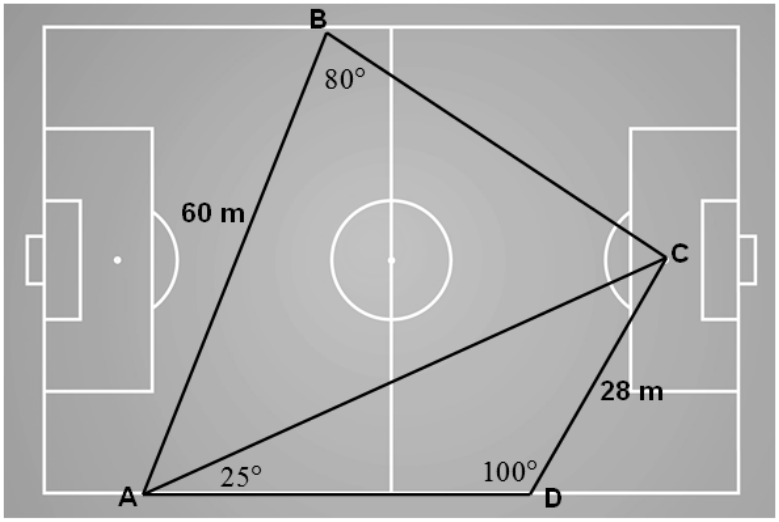
2.4 In the figure given below $PS \perp QR$; $\hat{Q} = 63°$; $S\hat{P}R = 49°$ and PQ=7,3 m.



Calculate the length of SR. (4)

2.5 Four of the world's greatest soccer players are practicing and are standing at points A, B, C and D on a soccer field.

AB = 60 m, CD = 28 m, $A\hat{D}C = 100°$, $C\hat{A}D = 25°$ and $A\hat{B}C = 80°$.



2.5.1 Calculate the distance AC. Round off your answer to the nearest metre. (3)

2.5.2 Use the cosine rule to calculate distance BC. (4)
[25]

QUESTION 3

3.1 The mass in kg of 24 learners in a grade 10 class is given below:

52	48	28	85	43	50	29	35
32	36	37	45	57	41	45	42
54	37	48	31	46	36	57	59

3.1.1 Determine the median mass of the group. (2)

3.1.2 Determine the interquartile range for the data provided. (5)

3.1.3 Determine the upper and lower fence values. (4)

3.1.4 Construct a box and whisker diagram for the above information, indicating any outliers. (3)

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3.2 A survey was carried out on 180 people to determine the distances that they travel to work every day. The table below is a summary of the information gathered.

Distance (s) (in km)	**Frequency(f_i)**	**Midpoint** (x_i)	$f_i \times x_i$	**< Cumulative Frequency**
$0 < s \leq 6$	5	3	15	5
$6 < s \leq 12$	13	9	117	18
$12 < s \leq 18$	25	15	(a)	(b)
$18 < s \leq 24$	41	21	(c)	(d)
$24 < s \leq 30$	20	27	(e)	(f)
$30 < s \leq 36$	32	33	(g)	(h)
$36 < s \leq 42$	26	39	(i)	(j)
$42 < s \leq 48$	18	45	(k)	(l)
TOTAL	180		$\sum f_i x_i$	

3.2.1 The first TWO rows of the above frequency distribution table have been completed for you. Complete the rest of the rows by writing only the answer next of the letter (a–l) in the ANSWER BOOK. (6)

Use the table you completed in QUESTION 3.2.1 to answer the questions.

3.2.2 Calculate the mean distance travelled by the group. (3)

3.2.3 Use the graph paper on the ADDENDUM (attached) to draw the ogive curve using the less than cumulative frequency and the upper class limit. (5)

3.2.4 Use the ogive curve you drew on the graph paper to determine the median distance of the group. Indicate on the ogive curve how you obtained the answer.

Write your EXAMINATION NUMBER in the space provided on the ADDENDUM, detach it from the question paper and hand it in with the ANSWER BOOK. (2)

[30]

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QUESTION 4

4.1 At the end of the financial year the treasurer of African Cycle Club compared the actual income and expenses figures with what the club had budgeted for at the beginning of the year. The table below is a summary of his findings.

Study the information and write only the answer next to the question number (4.1.1–4.1.6) in the ANSWER BOOK.

ITEM	BUDGETED AMOUNT	ACTUAL AMOUNT	VARIANCE
INCOME			
Membership fees	21 000	20 000	-1 000
Sponsorships	10 800	9 000	**(4.1.1)**
Fund-raising	**(4.1.2)**	7 600	+1 500
Total	37 900	36 600	-1 300
EXPENSES			
Rent	12 000	12 000	0
Water and electricity	1 975	**(4.1.3)**	+200
Telephone	670	910	-240
Subscription fees	**(4.1.4)**	7 200	0
Catering	4 300	5 100	-800
Prizes	7 500	8 200	**(4.1.5)**
Year-end function	2 500	2 600	-100
Total	36 145	37 785	-1 640
Surplus	**(4.1.6)**		

(6)

4.2 Mandla wants to buy a car for R210 000. The bank offers to finance the loan for a period of 5 years at an interest rate of 12% p.a. compounded monthly. His uncle offers to lend him the money which he must pay back after 5 years at an interest rate of 12,5% p.a. SIMPLE interest.

How much will Mandla save if he chooses the cheaper option? Show ALL working. (5)

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4.3 Veronica is a very thrifty person. She had R20 000 in the bank for which she earned an interest rate of 8,5% p.a. SIMPLE interest.

After 3 years she deposits a further R10 000 into the same account. The bank now offers her an interest rate of 10% p.a. interest compounded annually.

At the beginning of year 5 (2 years later) the bank increases the rate to 11,5% p.a. compounded semi-annually.

4.3.1 Draw a time line of the above investment. (3)

4.3.2 Determine the amount of money that Veronica has in the bank at the end of 7 years. (6)

[20]

TOTAL: 100

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FORMULA SHEET

1. Slant surface area of a pyramid = $\frac{1}{2}a\,l\,n$ or $\frac{1}{2}l\,h_s\,n$ (where n = number of sides)
2. Surface area of triangular pyramid = $\frac{1}{2}bh+\frac{1}{2}pl$ where p=perimeter of the base.
3. Surface area of a pyramid with an equilateral triangle as base = $\frac{\sqrt{3}}{4}s^2+\frac{1}{2}pl$
4. Surface area of an equilateral triangular pyramid = $4\times\frac{\sqrt{3}}{4}s^2$
5. Surface area of square pyramid = $b^2+\frac{1}{2}pl$
6. Surface area of a regular hexagonal pyramid = $\frac{3\sqrt{3}}{2}b^2+\frac{1}{2}pl$
7. Volume of a pyramid = $\frac{1}{3}$(area of base) $\times\perp$ height
8. $s=\frac{1}{2}(a+b+c)$ *and* a,b,c *are the sides of the triangle*
9. $A=\sqrt{s(s-a)(s-b)(s-c)}$
10. Circumference of circle = $2\pi r$
11. Area of curved surface of a cone = $\pi r\,l$ or $\pi r\,h_s$
12. Slant height of a cone = $l=\sqrt{h^2+r^2}$ or $h_s=\sqrt{\perp h^2+r^2}$
13. Volume of cone = $V_{cone}=\frac{1}{3}\pi r^2\times\perp h$
14. Area of a sphere = $A=4\pi r^2$
15. Volume of a sphere = $V=\frac{4}{3}\pi r^3$
16. $m=\frac{\Delta y}{\Delta x}=\frac{y_2-y_1}{x_2-x_1}$
17. $(x_m\,;y_m)=\left(\frac{x_1+x_2}{2}\,;\frac{y_1+y_2}{2}\right)$
18. $\theta=\tan^{-1}m$
19. Distance = $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$
20. $\frac{\sin\theta}{\cos\theta}=\tan\theta$
21. $\sin^2\theta+\cos^2\theta=1$
22. $\frac{\sin A}{a}=\frac{\sin B}{b}=\frac{\sin C}{c}$
23. $a^2=b^2+c^2-2bc\ \cos\hat{A}$

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24. $A=\frac{1}{2}ab\sin\hat{C}$

25. $\bar{x}=\frac{\sum_{i=1}^{n}x_i}{n}$

26. $\bar{x}=\frac{\sum f_i\,x_i}{n}$

27. $Q_{j\ position}=\frac{j}{4}(n+1)$

28. $Q_i=Q_3-Q_1$

29. Fence = $Q_3+1{,}5(Q_i)$

30. Fence = $Q_1-1{,}5(Q_i)$

31. $Me=l+\frac{\left(\frac{n}{2}-F\right)}{f}\times c$

32. $Mo=l+\frac{f_m-f_{m-1}}{2f_m-f_{m-1}-f_{m+1}}\times c$

33. $I=A_0\times\frac{r}{100}\times t$ or $I=\frac{P\,r\,t}{100}$ or $A_t=P(1+in)$

34. $A_t=A_0\left(1+\frac{r}{100\times m}\right)^{t\times m}$ or $A_t=P(1+i)^n$

35. $A_t=A_o(1-\frac{r}{100})^t$ or $A_t=P(1-i)^n$

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
DEPARTEMENT VAN HOËR ONDERWYS EN OPLEIDING

BOE 8/9

GRAPH PAPER • GRAFIEKPAPIER

(Return this sheet with the other answers)
(Lewer hierdie blad in saam met u antwoordboek)

Examination number
Eksamennommer

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